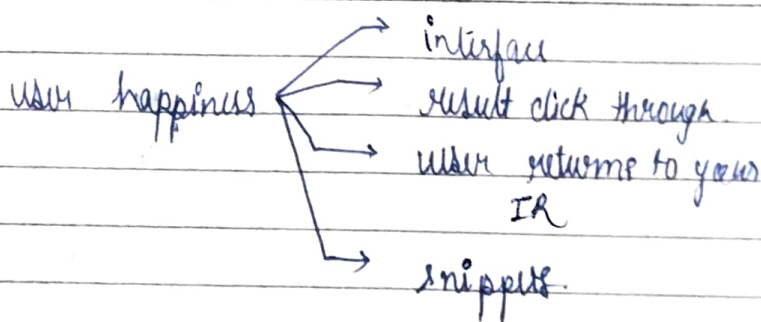


System Quality and User Utility:

System issues

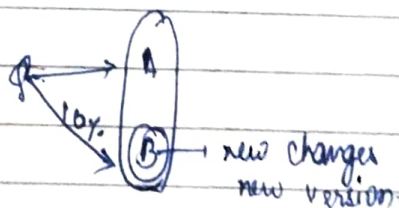
- ① How fast it index doc/hour
for a certain distribution of length.
- ② How fast is search.
latency as a function of index size
- ③ How expressive the query language.
- ④ How large is doc collection.

User utility:



Retiring a deployed system

A/B testing



10% Query to B : if B perform better it is
deployed and A retired.
★ ~~click~~ through log analysis. (measure to check)

Result snippet : short summary of document in search results to help users decide whether to click it.

Two types : 1. Static snippet : metadata, first 2 lines
2. Dynamic snippet :

- display one or more windows
- extract text around the query keywords
- Key word in context (KWIC)
- Highlight imp. parts.

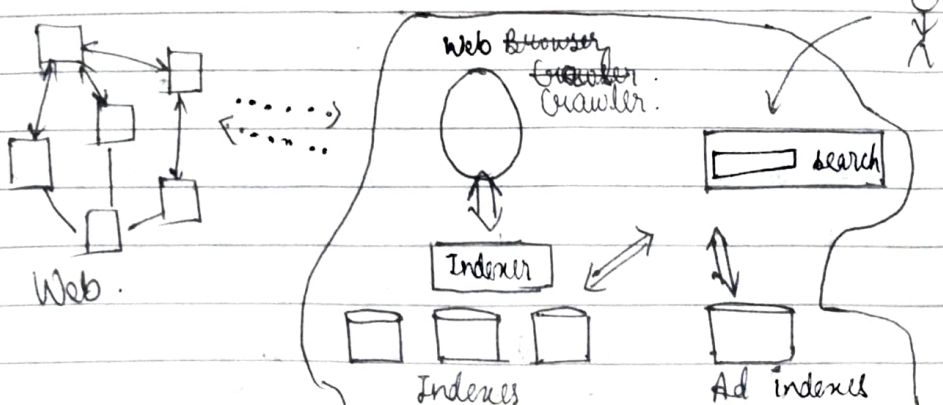
In dynamic we have to dynamically compute
* this should be fast (this is also imp aspect).

Result snippet : given a variety of keyword occurrences in a document, the goal is to choose fragments which are :

Dynamic

- maximally informative above the discussion of the term in document.
- self contained enough to be easy to read.
- short enough to fit within the normally strict constraints on the space available for summaries.

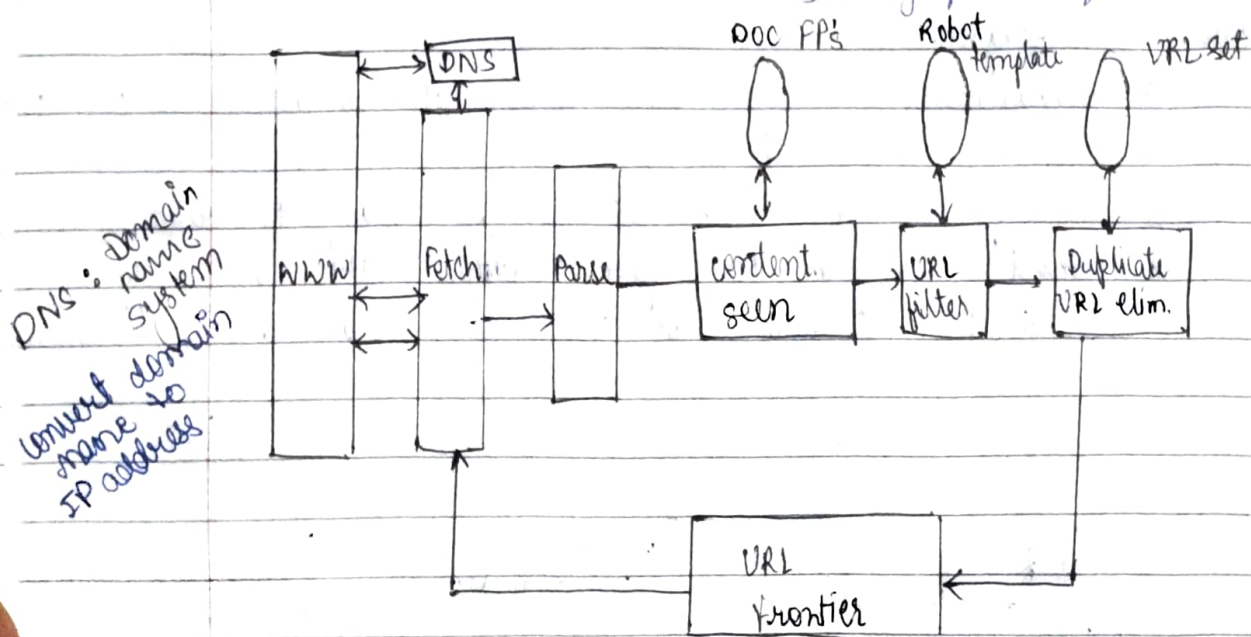
Various components of Web search engine :



- Crawler collects data from web.
- Indexer organizes into indexes.
- User enters query in search box.
- Search engine retrieves and shows results.

Features of web crawler:

- ① Robustness (spider trap handle)
- ② Politeness (rule for crawling rate)
- ③ Distributed (divided across multiple machines)
- ④ Scalable (grow easily to handle huge no. of web pages)
- ⑤ Performance and Efficiency (quickly while minimal resource)
- ⑥ Quality (good quality web pages, useful)
- ⑦ Freshness (rate at which change how often crawler has)
- ⑧ Extensible (page protocols etc. are already made) (ready for new formats)



It shows working of web crawler. Starts with URL frontier, fetches web pages using DNS and parse their content to extract links. Then it filters and remove duplicate URLs before adding new links back to frontier for continuous crawling.

Basic properties of any non professional crawler should satisfy:

- Only one connection should be open to any given host at a time.
- A waiting time of few second should occur between successive requests.
- Politeness restrictions should be obeyed.